

Final Project Part 3 Assignment

1. Project Motivation/Introduction

As soon as I first landed in this city, I realized living in Reno was not going to be easy because of the very dry air, especially for a man who was born and raised in Japan, where it is always humid. Low humidity harmed my health by causing dry skin, dry eyes, a sore throat, and more. It can also make it harder to focus on one thing since you would want to drink water more often, increasing the number of times you have to go to the bathroom. Additionally, many people do not realize how dry their environment is until they begin to feel these effects. For these reasons, my plan is to create a small system that can monitor humidity levels of the air and give a warning using multiple LEDs when the air becomes too dry.

An embedded system would be a good solution because it can run continuously with very high power efficiency. Instead of using phone apps or guessing the room condition, users can easily understand the environment just by observing the color of the LED lights. The main goal of this project is to build something practical, affordable, and easy to use in everyday life while making sure the system will not be too complex.

2. Proposed System Overview

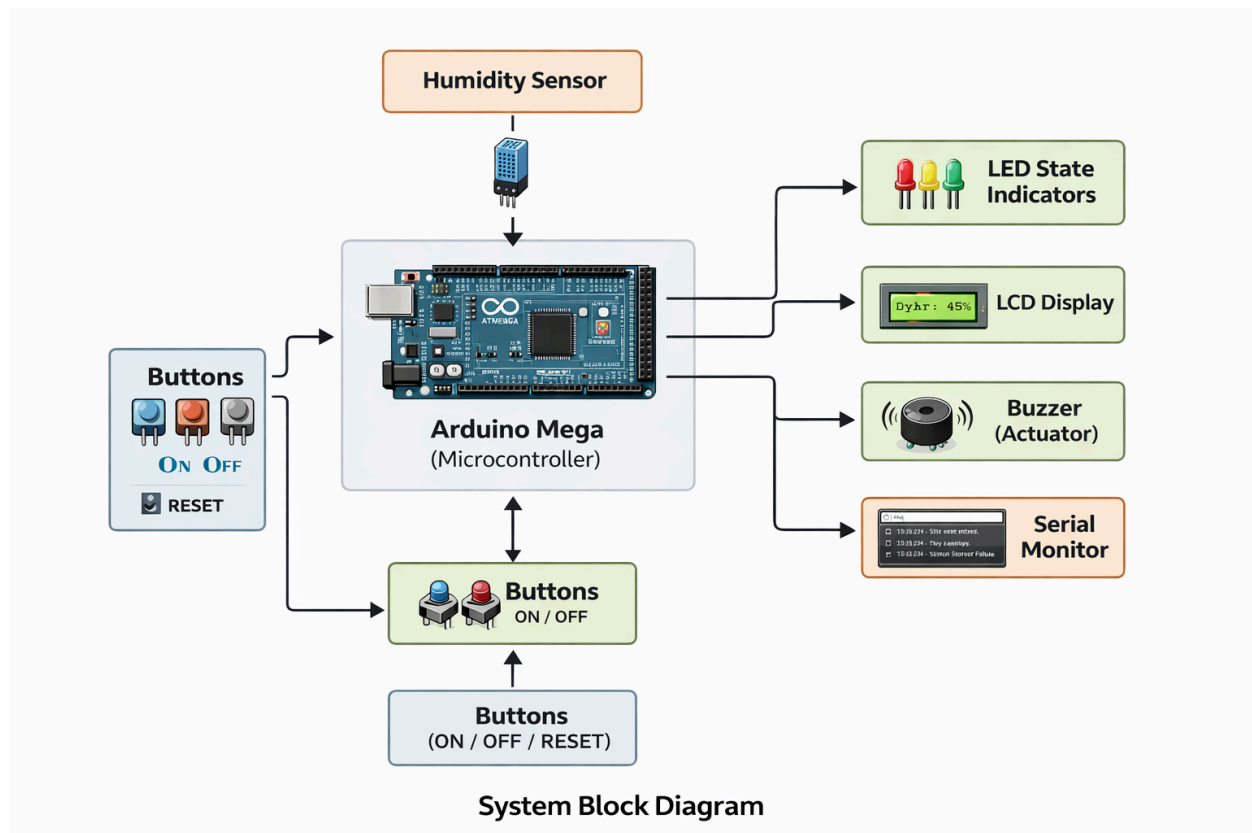
The proposed system is a compact humidity monitoring device that is built using the ELEGOO Mega R3 starter kit. The system uses a potentiometer to simulate humidity sensor values through ADC input. The Arduino microcontroller processes data passed by the sensor and gives outputs including LEDs, a buzzer, and an LCD display.

The system automatically begins the operation once it is turned on. It repeatedly reads sensor data and updates the system state according to the conditions defined beforehand. LEDs are used to indicate the current system state. The LCDs display both the humidity value and the current

state so that the user can easily understand the condition. A buzzer is used as an actuator to alert the user when the air becomes too dry.

The users can control the system by using three buttons: ON, OFF, and RESET. The ON button starts the system and moves it from OFF to IDLE. The OFF button stops the system from any state. The RESET button can reset the system from the ERROR state. Overall, the system focuses on simple operation, clear feedback, and real-time monitoring of indoor air conditions.

2.1. System Block Diagram



In the actual circuit, a potentiometer is used instead of a humidity sensor for easier demonstration.

3. System Behavior (High-Level)

When the system is turned on, it goes into the IDLE state, then the system begins to check the humidity level. The humidity sensor sends data to the microcontroller, and the LCDs display the

humidity level. In this state, the actuator is not working because the system is just checking the environment. The IDLE LED is turned on to show that the system is working properly.

If the humidity level gets too low, the system changes from IDLE to ACTIVE. When the system is ACTIVE, the buzzer makes a sound to inform the user that the air is too dry. The LCD shows the humidity level and a warning message. The ACTIVE LED is also on. If the humidity level gets back to normal, the system goes back to IDLE, and the buzzer stops making a sound.

If the sensor reading is not working correctly, the system goes into the ERROR state. In this state, the buzzer stops making a sound, the ERROR LED is turned on, and the LCD shows an error message. The system stays in the ERROR state until the RESET button is pressed. The OFF button can be pressed at any time to turn the system off. Once it is pressed, the system stops checking the environment and turns off the actuator.

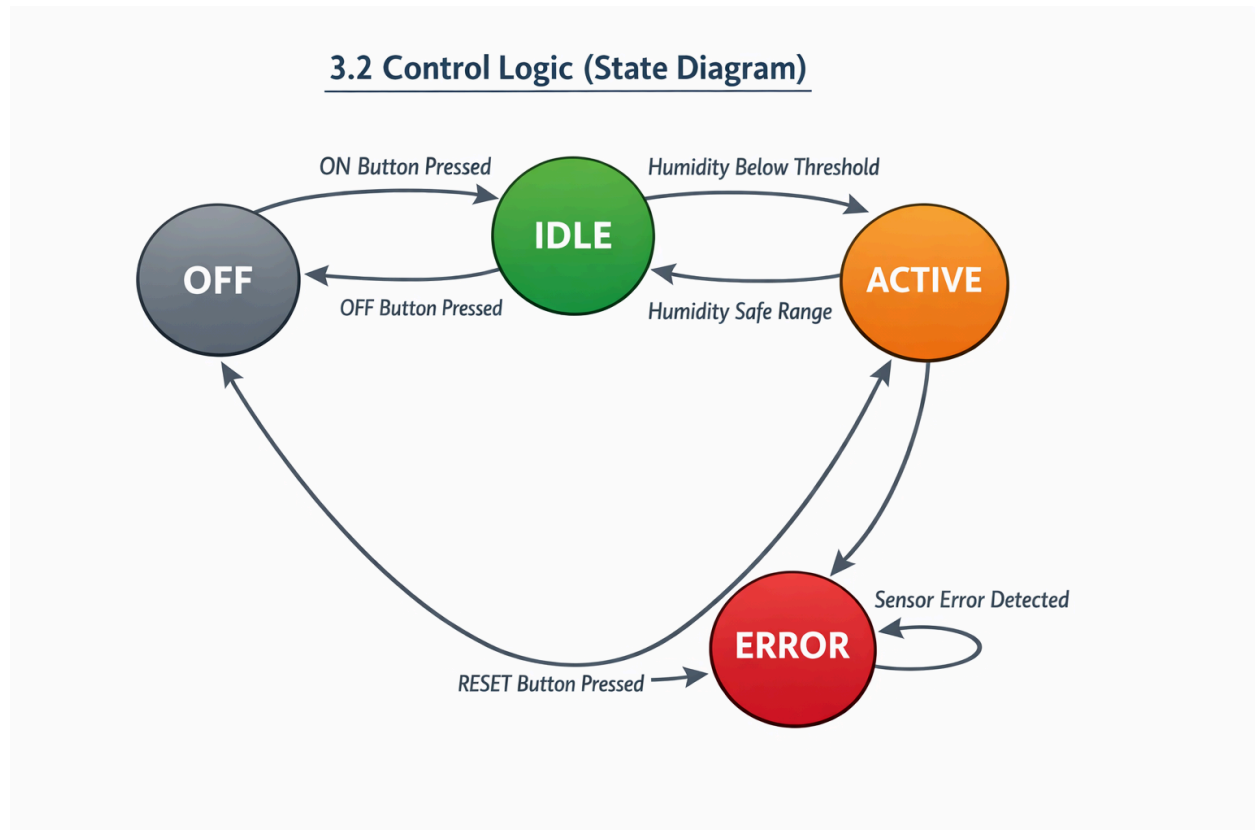
3.1. Module Design

The Sensor Module reads humidity data from the sensor and gives real-time environmental data to the system. The data is used to determine if the air is safe or not.

- A. The Control Logic Module: This module mainly manages system states. There are four states: OFF, IDLE, ACTIVE, and ERROR. It decides what the system needs to do based on sensor data and button inputs. It makes sure that the system responds correctly.
- B. The Display Module: This module updates the LCD screen. It shows the humidity value and system state. This helps the user understand what the system is doing.
- C. The Actuator Module: This module controls the buzzer. When the air is too dry, the buzzer is turned on. In certain conditions, the actuator is turned off. It is disabled in the ERROR state.

- D. The Button/Input Module: This module handles user input. This input comes from the ON, OFF, and RESET buttons. The ON button uses an interrupt. This starts the system quickly. The OFF and RESET buttons control system behavior and recovery.

3.2 Control Logic (State Diagram or Flowchart)



3.3 Environmental & Energy Impact

This project helps people use energy only when they really need to, promoting energy efficiency. For example, if you use this system, it will tell you when the air in your room is too dry. This helps you decide when to turn the humidifier on, so you do not waste energy when you do not need to.

Using the parts from the ELEGOO Mega R3 kit ensures safer indoor use and reduces the risk of electrical hazards, as they can only work under low voltage. Also, the system is not too

complicated, and it does not consume much energy because the actuator only turns on when it needs to.

This project supports sustainability because it uses parts that are affordable and reusable. The system is easy to build and fix, reducing the waste and supporting long-term use. The system is also accessible to anyone, even without technological knowledge, since the interface is kept simple and easily understandable.

4. Assumptions & Constraints

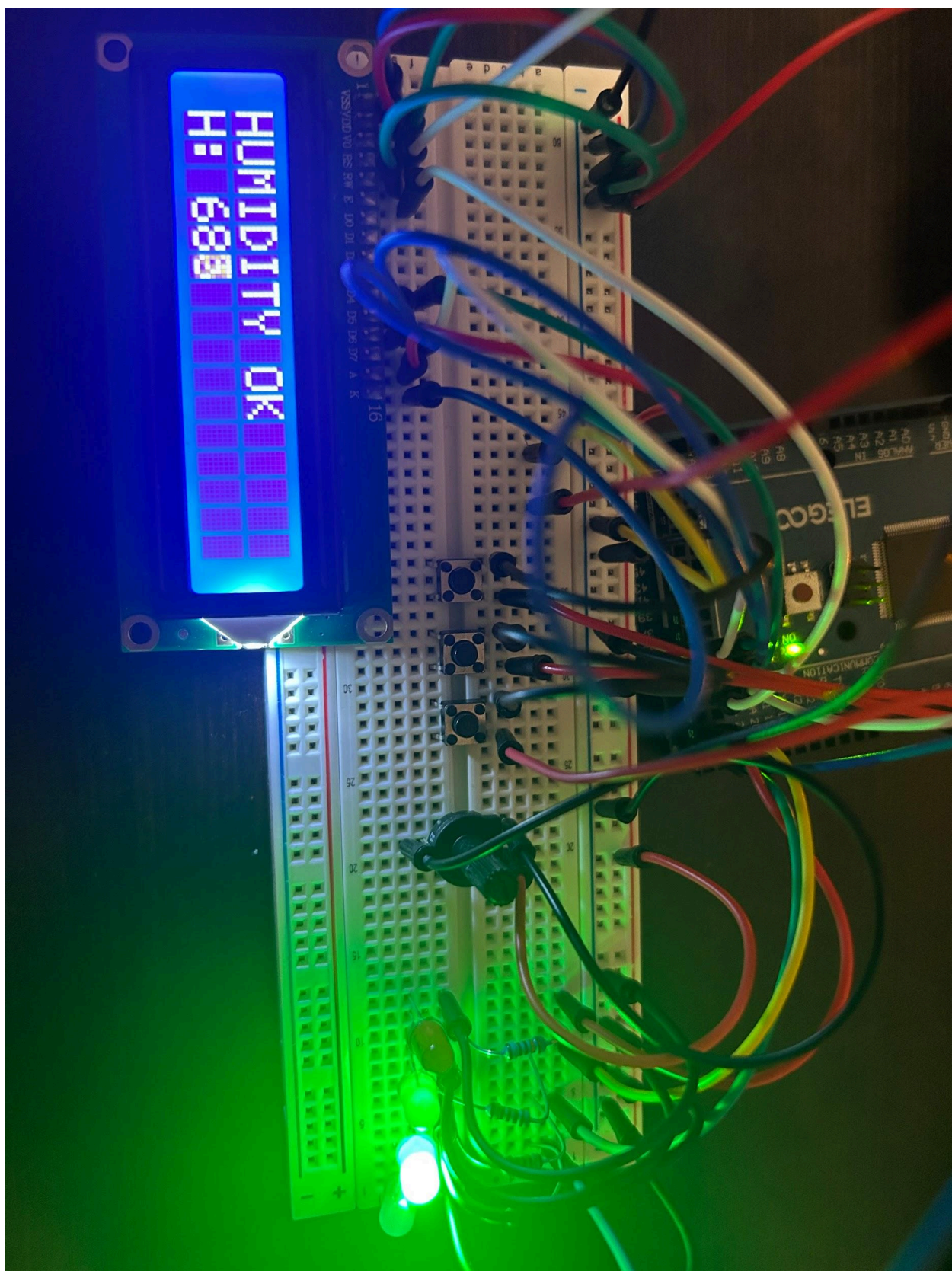
This project only uses parts from the ELEGOO Mega R3 Ultimate Starter Kit. The system is also a prototype that focuses on simple functions. Thus, it does not support the industrial-level performance.

Additionally, the microcontroller has limited power and memory. This limits the system so that it can only work under a simple logic and efficient design. The time frame of the semester is also a constraint, which means the system can't be too complicated.

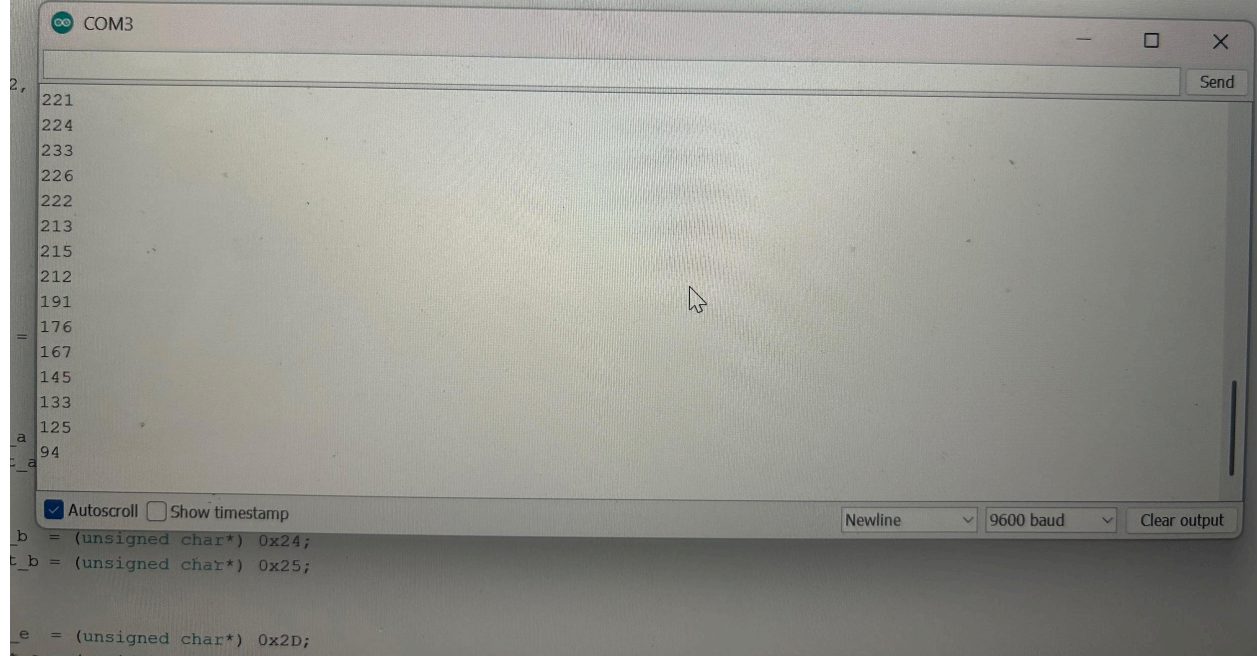
Also, the system is only meant for small indoor spaces like a bedroom or study area. It might not work properly in large rooms or environments where the conditions change quickly and unexpectedly.

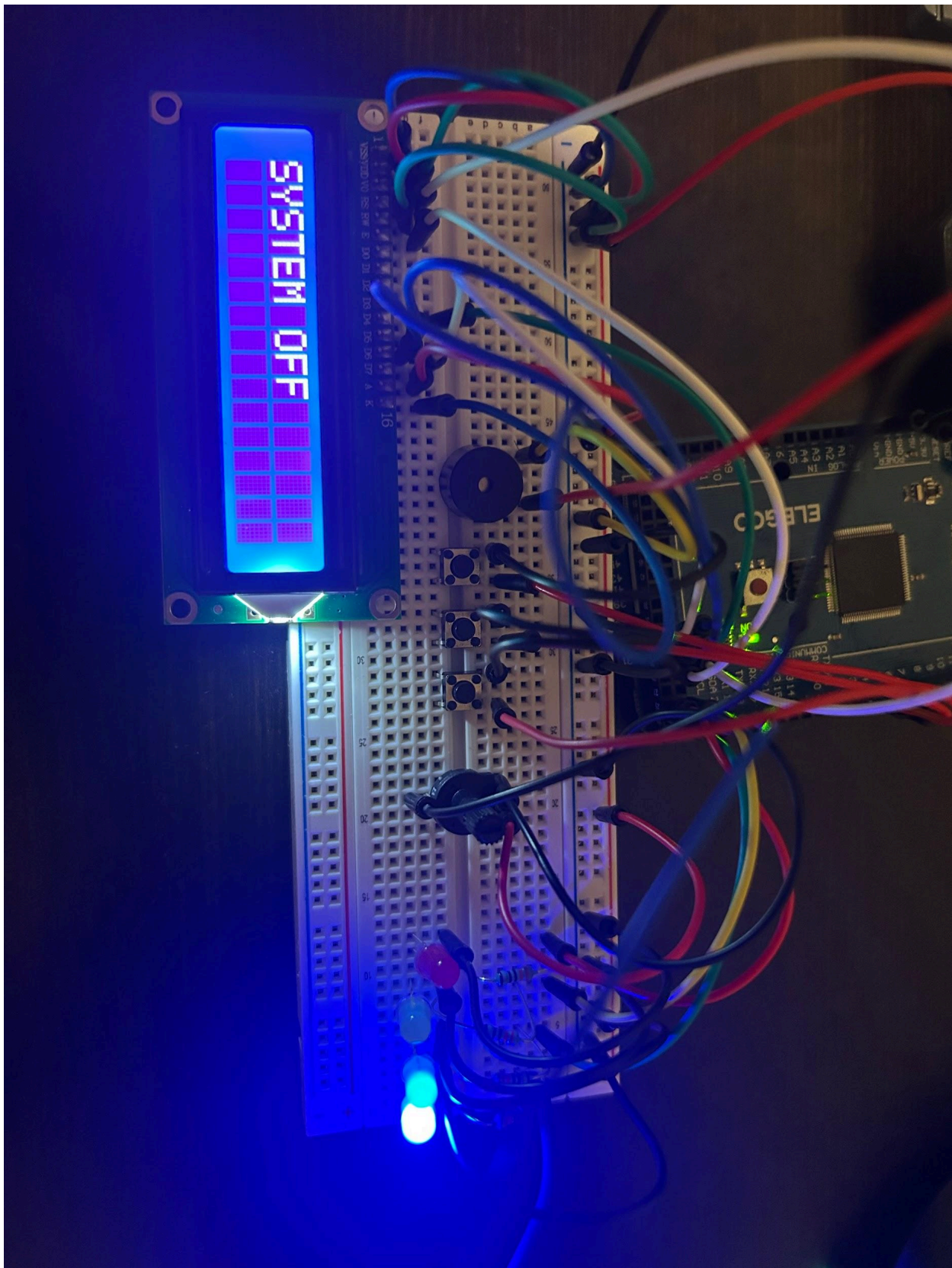
Another concern would be that the sensor readings can be affected by noise or the placement, which can affect the accuracy of the system.

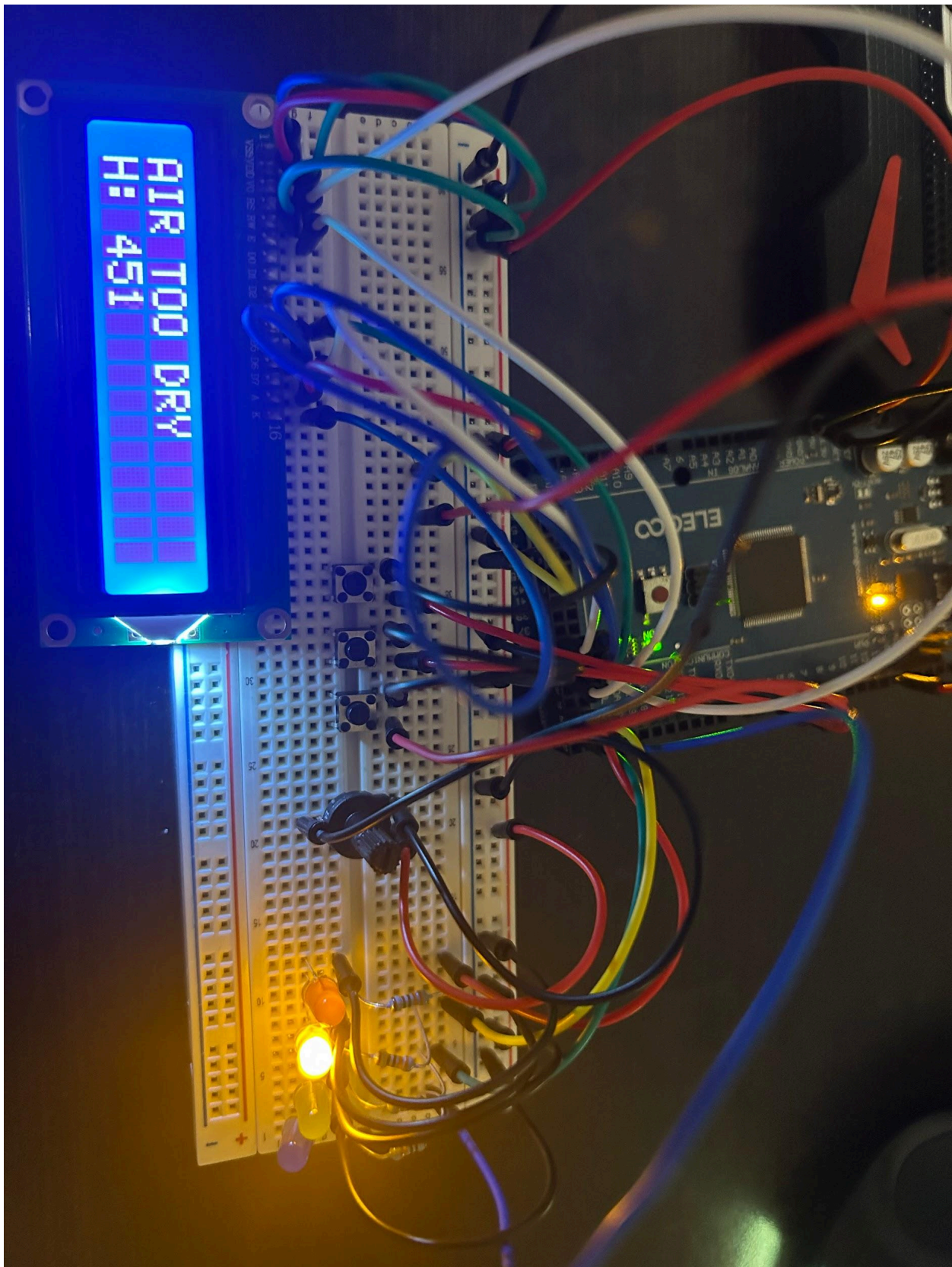
5. Result

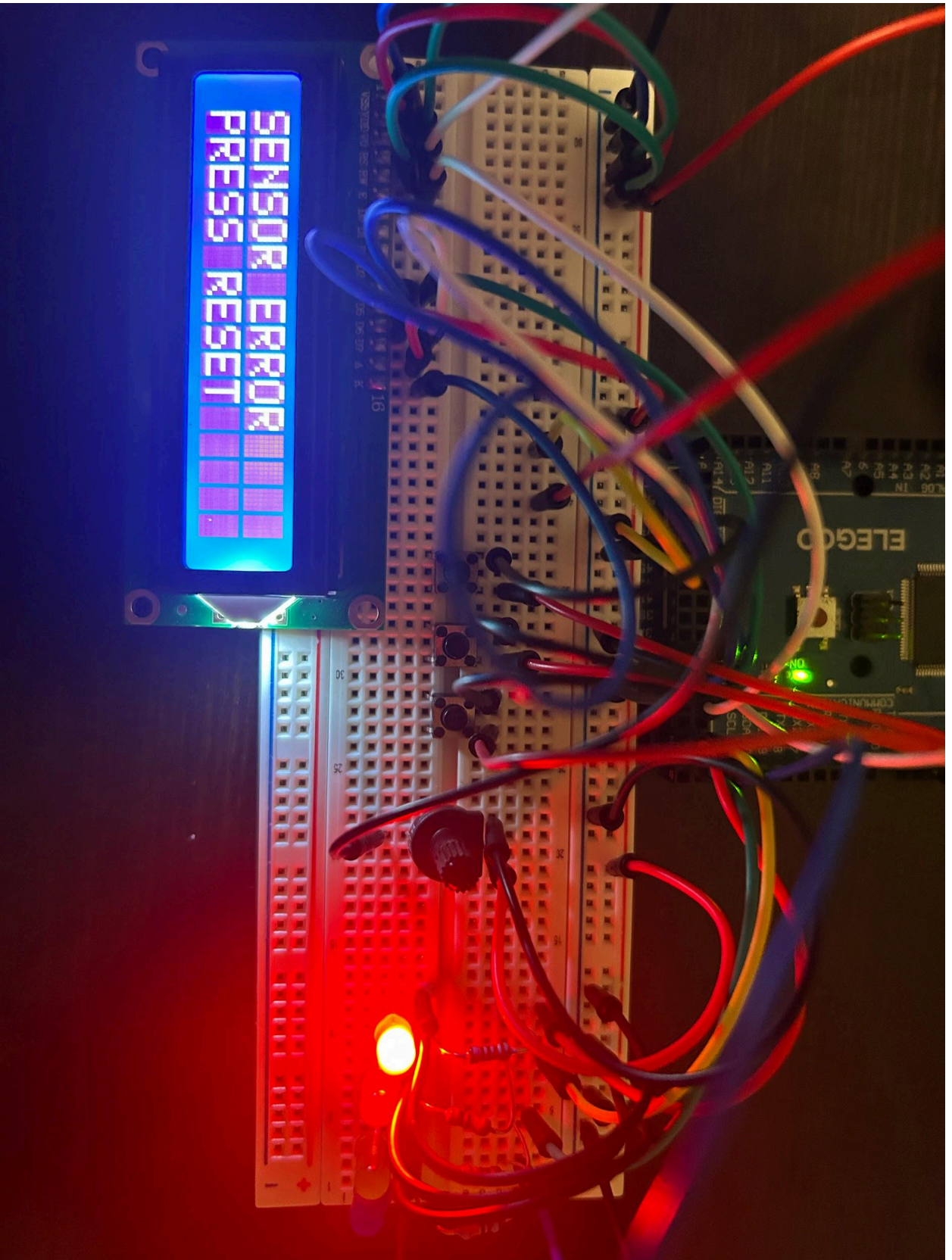


(Windows Store 1.8.57.0)









6. Conclusion

The project successfully demonstrated a humidity monitoring system using register-level programming and a state machine design. The LEDs, LCD, buzzer, buttons, and ADC input all worked properly. One limitation is that the system uses a potentiometer to simulate humidity values instead of a humidity sensor. In the future, the project could be improved by adding a real sensor and more features.